

BORONG XU

Computer Engineering Student

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boring180

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ABOUT ME

- Undergraduate Computer Engineering student at HKUST, passionate about robotics and embodied intelligence. Skilled in C/C++, Python, and ROS with hands-on project experience. Eager to contribute to intelligent robot systems in real-world applications.

EDUCATION

The Hong Kong University of Science and Technology (HKUST)

BACHELOR OF ENGINEERING IN COMPUTER ENGINEERING

Hong Kong SAR

Graduated: June 2026

- GPA 3.84/4.3 (Ranked 1/98 in program)
- Major GPA 4.03/4.3

University of New South Wales (UNSW Sydney)

VISITING STUDENT (EXCHANGE), COMPUTER SCIENCE AND ENGINEERING

Sydney, Australia

Feb 2025 - May 2025

High School Affiliated to Renmin University of China (RDFZ)

HIGH SCHOOL DIPLOMA(GAOKAO)

Beijing, China

Graduated June 2022

- Activities:
 - FIRST Robotics Club, Chief Engineer (season 2021)
 - FIRST Robotics Club, Alumni Mentor (season 2022-2024)

PROJECT EXPERIENCE

LLM Agent Development

UNDERGRADUATE RESEARCH OPPORTUNITY (UROP)

Supervised by Prof. Shenghui Song

Jun 2024 - May 2026

- Designed and implemented a Large Language Model (LLM) agent capable of playing board games. Conducted comprehensive evaluations of agent strategies and performance.

ELEC5660 Introduction to Aerial Robotics

COURSE PROJECT

Supervised by Prof. Shaojie Shen

Feb 2026 - May 2026

- Developed aerial robot algorithms from scratch, including visual-inertial odometry, Extended Kalman Filter (EKF) state estimation, and autonomous path planning within a simulation environment, enabling robust UAV localization, mapping, and navigation.

Quadruped Robot Control

RESEARCH PRACTICUM, UNSW SYDNEY

Supervised by Prof. Maurice Pagnucco and Prof. Yang Song

Feb 2025 - Aug 2025

- Built and programmed a Unitree GO2 quadruped robot to perform complex tasks, integrating depth camera for accurate object localization (localization error < 10cm) and LiDAR for SLAM.

Underwater Localization System

FINAL YEAR PROJECT

Supervised by Prof. Huan Yin and Prof. Fumin Zhang

Jun 2025 - May 2026

- Developed a Multi-Camera Vision-Based Localization System for Tracking Multiple Autonomous Underwater Vehicles (AUVs). Reduced localization error in complex underwater lighting scenarios, improving positioning accuracy to within 10cm in a 1m x 1m range.

HONORS

2022-2026 **Dean's List**, 5 times

HKUST

2023-2026 **Continuing Undergraduate Students Scholarship**, 10,000 HKD per year

HKUST

2023 **Bronze Medal**, The Innovation Competition: Our Livable City, We Engineer

HKIE

2022 **Entry Scholarship**, 70,000 HKD

HKUST



SKILLS

- C/C++:** Eigen, PCL, STL
- Python:** PyTorch, OpenCV, Numpy
- ROS:** Navigation2, Gazebo, Slam Toolbox, TF

LANGUAGES

English • Fluent (IELTS 7.5)
Mandarin • Mother tongue